# Introduction to Electronics

08.10.2021

Assoc. Prof. Engin AFACAN

# Courses...

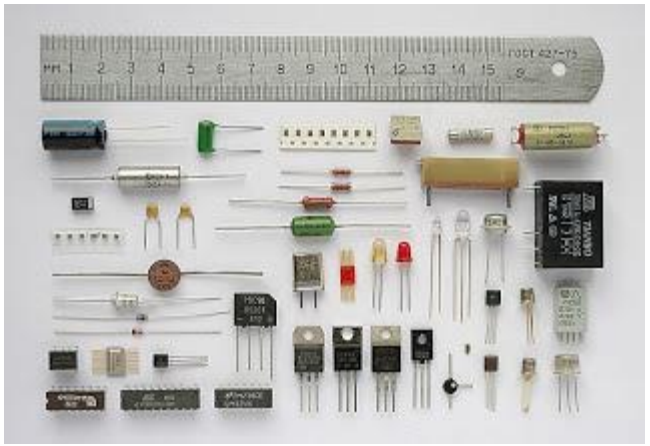
| Lisans Ders Kataloğu | | | | | | | |
|---|---|---|---|---|---|---|---|
| Ders Kodu | Ders Adı | Öğretim Dili | Dönemi | Ders Tipi | GTÜ Kredisi | T+U+L | AKTS Kredisi |
| ELM 101 | Elektronik Mühendisliğine Giriş ve Kariyer Planlama | Türkçe | Güz | Zorunlu | 2 | 2+0+0 | 2 |
| ELM 218 | Olasılık ve Rastgelelik | Türkçe | Bahar | Zorunlu | 3 | 3+0+0 | 5 |
| ELM 225 | Devre Teorisi I | İngilizce | Güz | Zorunlu | 3 | | 4 |
| ELM 226 | Elektrik Devreleri Laboratuvarı I | İngilizce | Güz | Zorunlu | 1 | | 2 |
| ELM 227 | Devre Teorisi II | İngilizce | Bahar | Zorunlu | 3 | | 4 |
| ELM 228 | Elektrik Devreleri Laboratuvarı II | İngilizce | Bahar | Zorunlu | 1 | | 1 |
| ELM 231 | Elektronik Elemanları | Türkçe | Güz | Zorunlu | 3 | 3+0+0 | 5 |
| ELM 234 | Lojik Devreler ve Tasarım | Türkçe | Bahar | Zorunlu | 3 | | 4 |
| ELM 235 | Lojik Devre Tasarım Laboratuvarı | Türkçe | Bahar | Zorunlu | 1 | | 2 |
| ELM 236 | Elektronik I | İngilizce | Bahar | Zorunlu | 3 | | 4 |
| ELM 237 | Elektronik I Laboratuvarı | İngilizce | Bahar | Zorunlu | 1 | | 1 |
| ELM 264 | İşaretler ve Sistemler | Türkçe | Bahar | Zorunlu | 3 | 3+0+0 | 5 |
| ELM 321 | Lojik Devreler ve Tasarım | Türkçe | Güz | Zorunlu | 4 | 4+0+0 | 6 |
| ELM 322 | Kontrol Sistemleri | İngilizce | Bahar | Zorunlu | 4 | 3+0+0 | 5 |
| ELM 323 | Elektromekanik Enerji Dönüşümü | İngilizce | Güz | Zorunlu | 4 | 3+0+0 | 5 |
| ELM 334 | Mikro İşlemciler | İngilizce | Güz | Zorunlu | 3 | | 4 |
| ELM 335 | Mikroişlemci Laboratuvarı | İngilizce | Güz | Zorunlu | 1 | | 2 |
| ELM 336 | Elektronik II | İngilizce | Güz | Zorunlu | 3 | | 4 |
| ELM 337 | Elektronik II Laboratuvarı | İngilizce | Güz | Zorunlu | 1 | | 1 |
| ELM 341 | Elektromagnetik Alan Teorisi | Türkçe | Güz | Zorunlu | 3 | 2+2+0 | 5 |
| ELM 342 | Elektromanyetik Dalga Teorisi | Türkçe | Bahar | Zorunlu | 3 | 3+0+0 | 5 |
| ELM 361 | Analog Haberleşme Sistemleri | İngilizce | Güz | Zorunlu | 3 | 3+0+0 | 5 |
| ELM 365 | Sayısal Haberleşmenin Temelleri | İngilizce | Bahar | Zorunlu | 3 | | 5 |

# Courses

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| ELM 368 | Sayısal Işaret İşleme Laboratuvarı | Türkçe | Bahar | Zorunlu | 1 | | 2 |
| ELM 416 | Elektromanyetikte Nümerik Yöntemler | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 418 | Mühendislik ve Temel Bilimler İçin C++ | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | 3+0+0 | 6 |
| ELM 421 | Sensörler ve Aktüatörler | İngilizce | Bahar | Alan seçmeli | 3 | | 6 |
| ELM 422 | Endüstriyel Enstrümantasyon ve Otomasyona Giriş | İngilizce | Bahar | Alan seçmeli | 3 | | 6 |
| ELM 425 | Güç Elektroniği | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | 3+0+0 | 6 |
| ELM 426 | Akıllı Elektrik Şebekeleri ve Güç Sistemlerine Giriş | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | 3+0+0 | 6 |
| ELM 429 | Sayısal Kontrol Sistemleri | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 431 | Kriptolojiye Giriş | İngilizce | Güz | Alan seçmeli | 3 | 3+0+0 | 6 |
| ELM 432 | Elektronik Devre Analizinde Özel Konular | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 433 | Numerik Hesaplama Yazilimlari | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 437 | Endüstriyel Kontrole Giriş | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 441 | Mikrodalga Teknikleri ve Cihazları | İngilizce | Güz ve Bahar | Alan seçmeli | 4 | 3+0+0 | 6 |
| ELM 442 | Radar Temelleri | Türkçe | Güz ve Bahar | Alan seçmeli | 3 | 3+0+0 | 6 |
| ELM 444 | Anten Teorisi | İngilizce | Güz ve Bahar | Alan seçmeli | 4 | 3+0+0 | 6 |
| ELM 446 | Elektromanyetik Uyumluluğun Temelleri | Türkçe | Bahar | Alan seçmeli | 3 | 3+0+0 | 6 |
| ELM 451 | Sayısal Tümleşik Devrelere Giriş | İngilizce | Güz | Alan seçmeli | 3 | | 6 |
| ELM 452 | Analog Tümdevreler | Türkçe | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 453 | Doğrusal Olmayan Elektronik Devreler | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 454 | Elektronik Devrelerin Simülasyonu | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 456 | İşlemsel Kuvvetlendiriciler ve Uygulamaları | İngilizce | Güz ve Bahar | Alan seçmeli | 3 | | 6 |
| ELM 457 | FPGA-tabanlı Sistem Tasarımı | İngilizce | Güz | Alan seçmeli | 3 | | 6 |
| ELM 458 | Gömülü Sistem Tasarımı | İngilizce | Bahar | Alan seçmeli | 3 | | 6 |
| ELM 461 | Haberleşme Teknikleri | Türkçe | Güz ve Bahar | Alan seçmeli | 3 | | 6 |

# Electronics???

- **Electronics:** is a field of science that examines systems carrying, processing and storing data using electricity.



- **Electronics Components :** Conductors, insulators, semi-conductors, resistor, inductor, capacitor, diode, transistor and other nano-scale components.



- **Electronics Engineering:** production of these components or design of circuits using them.

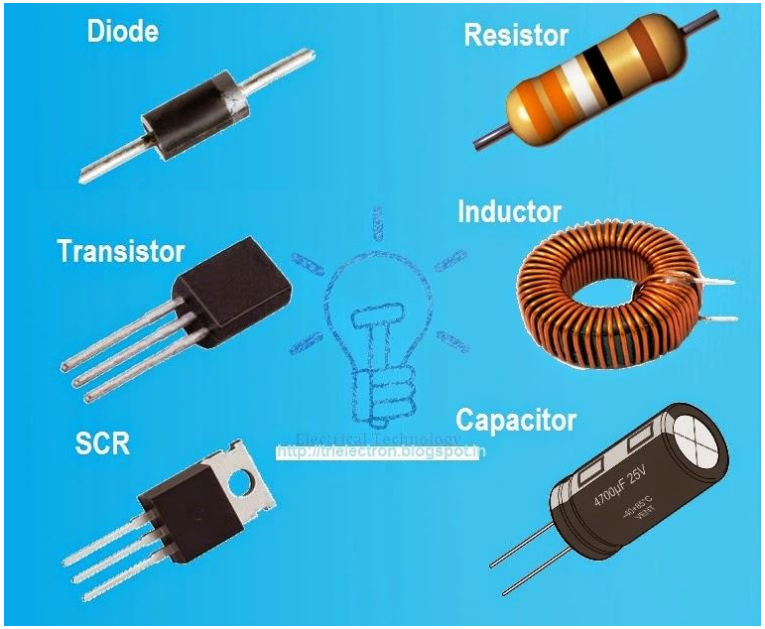
# Electronics Components

## Active Components

- Diode (D)
- BJT transistors (Q)
- FET transistors (M)
- SCR

## Passive Components

- Resistor (R)
- Capacitor(C)
- Inductor(L)

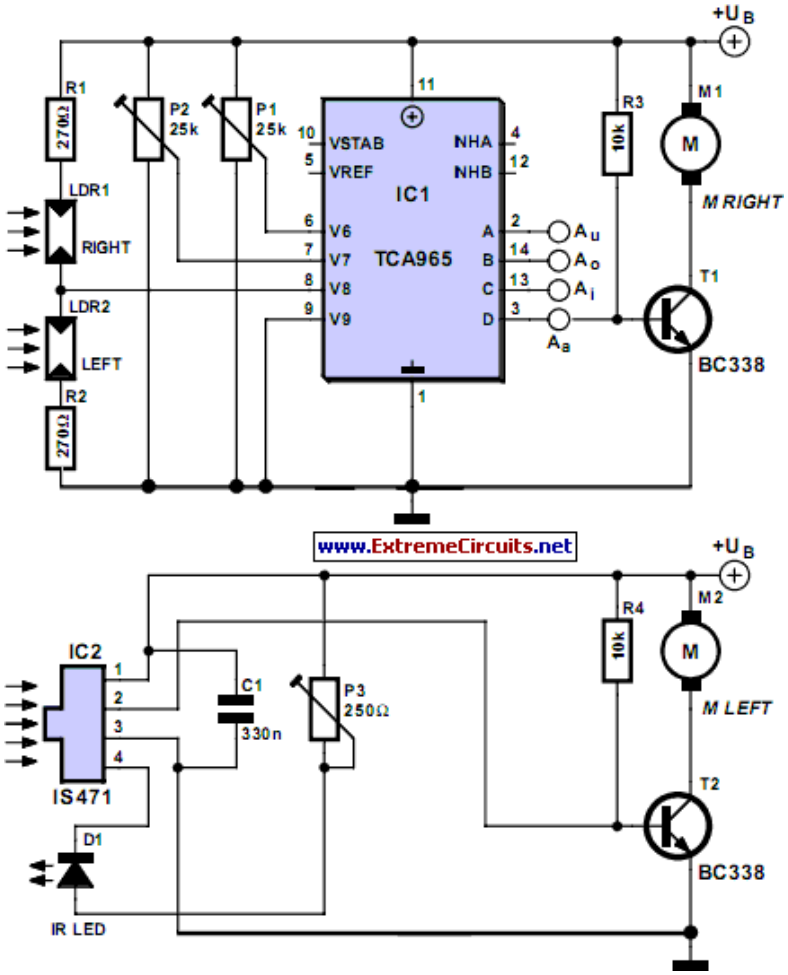


# Electronic Circuit Design

- Discrete Circuit Design (PCB)***
- Integrated Circuit Design (IC)***
- Programmable Circuit Design (FPGA)***
- MicroElectroMechanicalSystems(MEMs)?
- 2D/3D Integration (IC+MEMs)???

# Discrete Circuit Design

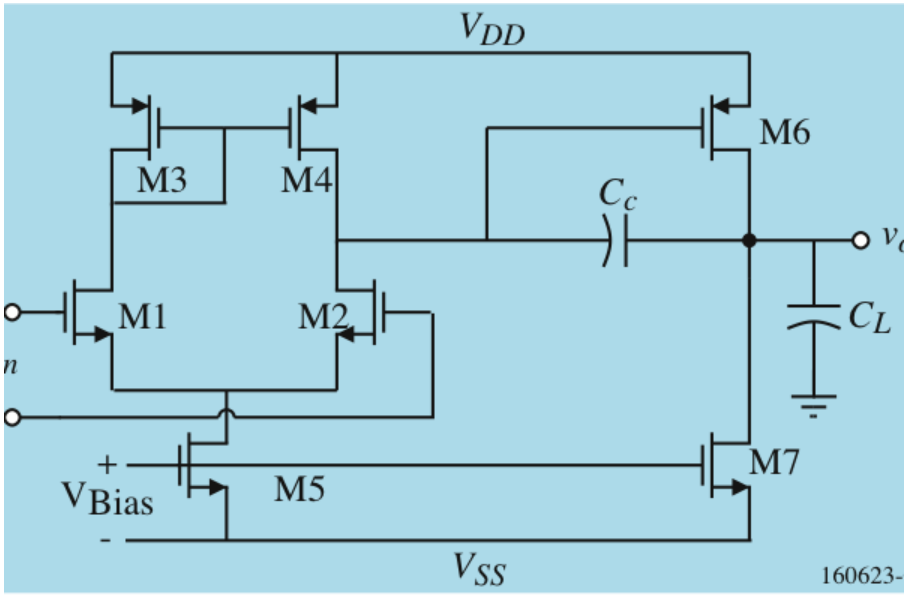
## Analog/Digital/Mixed/RF

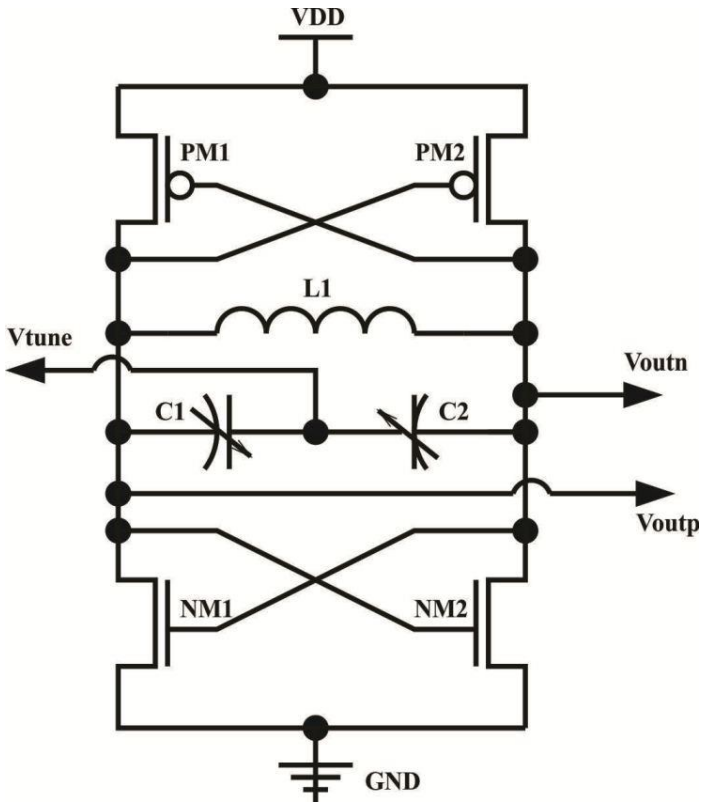





# Integrated Circuit Design-1

Analog/RF IC



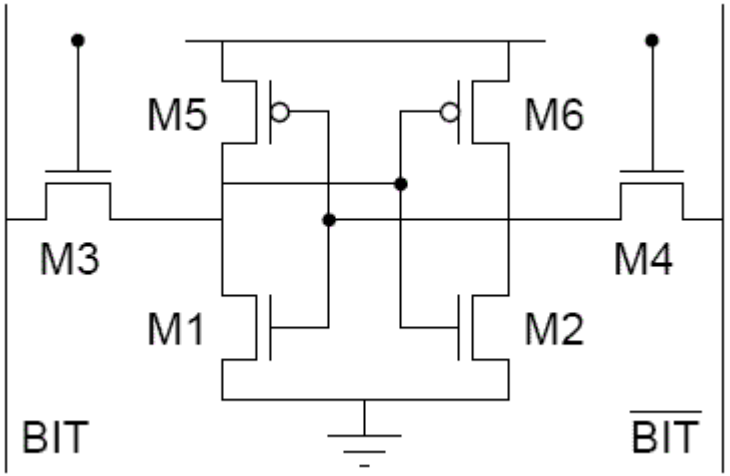



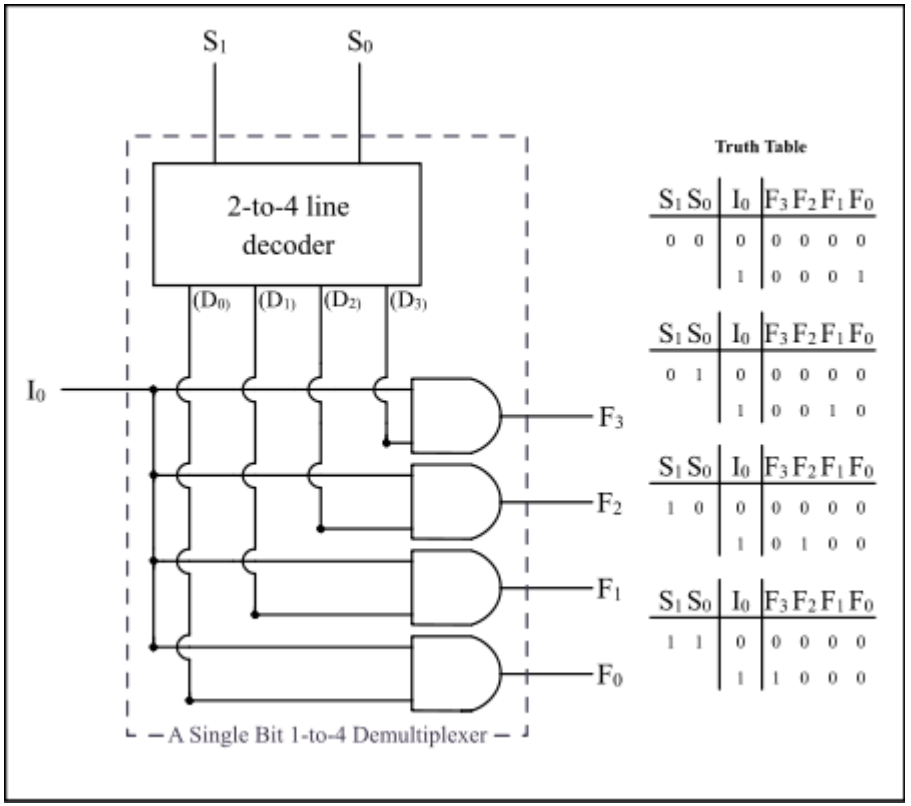

Analog→Amplifiers, comparators, etc.

RF→Filters, oscillators, LNA, Transmitter-Receiver etc.

# Integrated Circuit Design-2

Digital IC

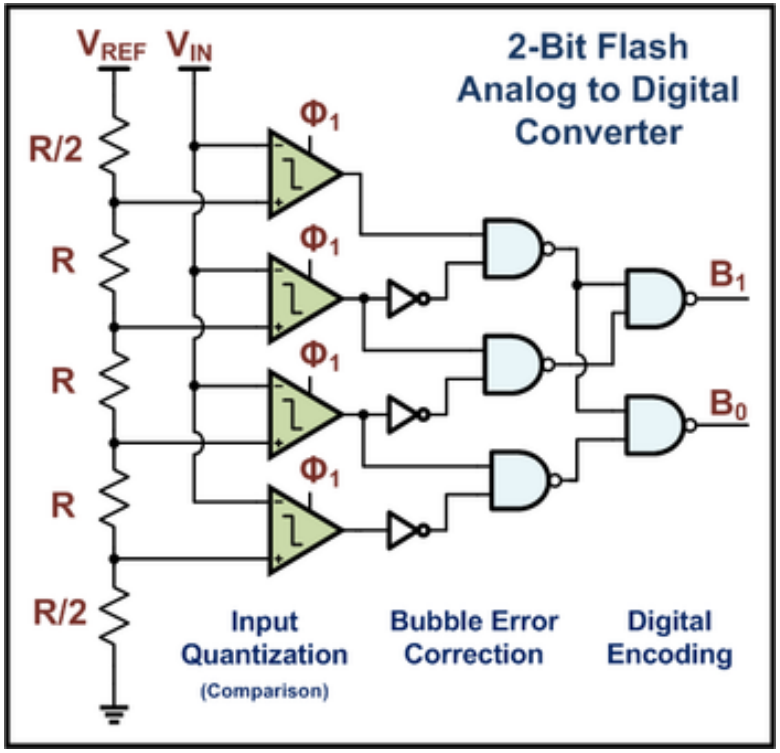



Digital→Counter, sRAM, multiplexer, digital signal process etc.

# Integrated Circuit Design-3

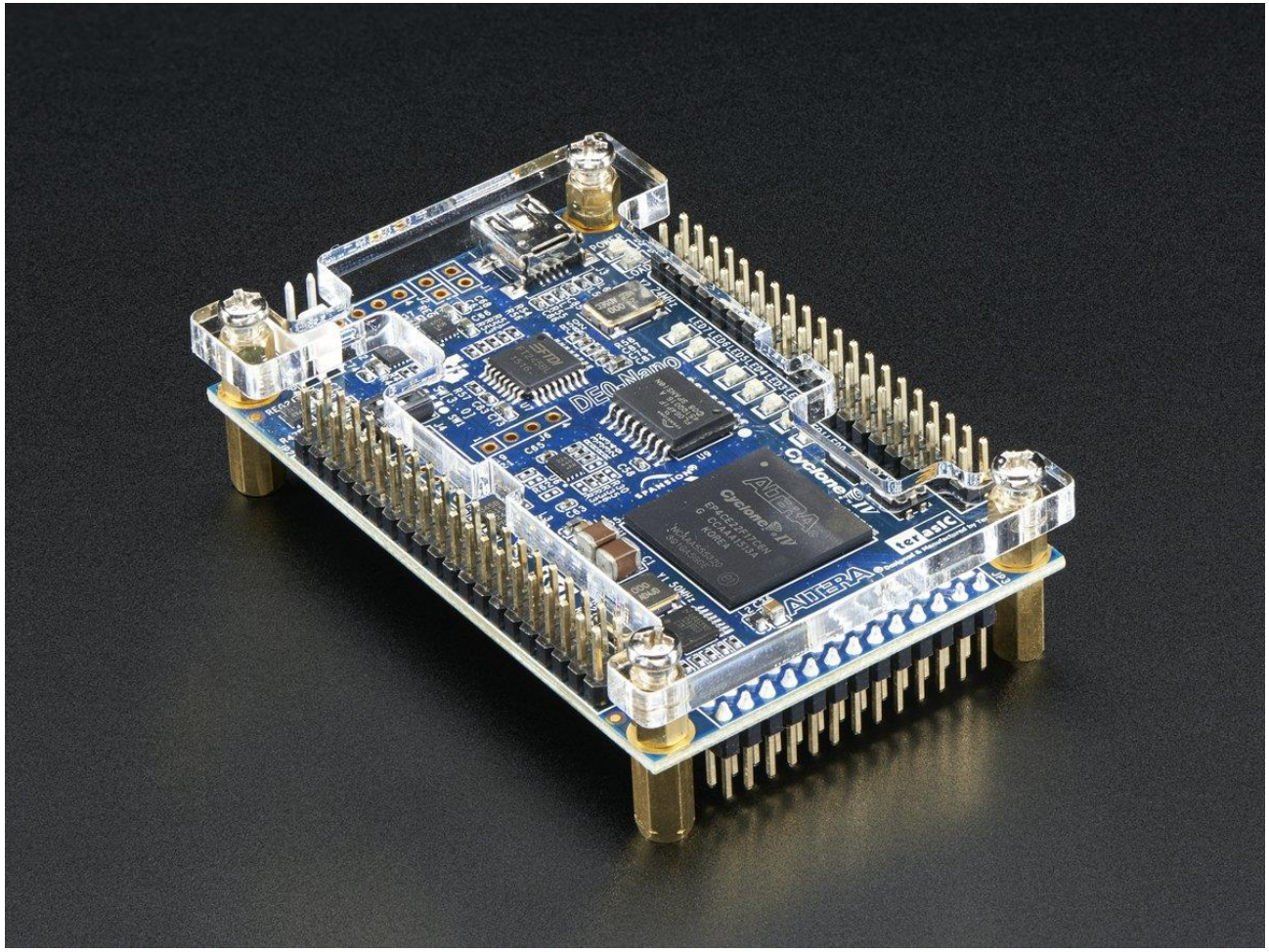
Mixed Signal IC



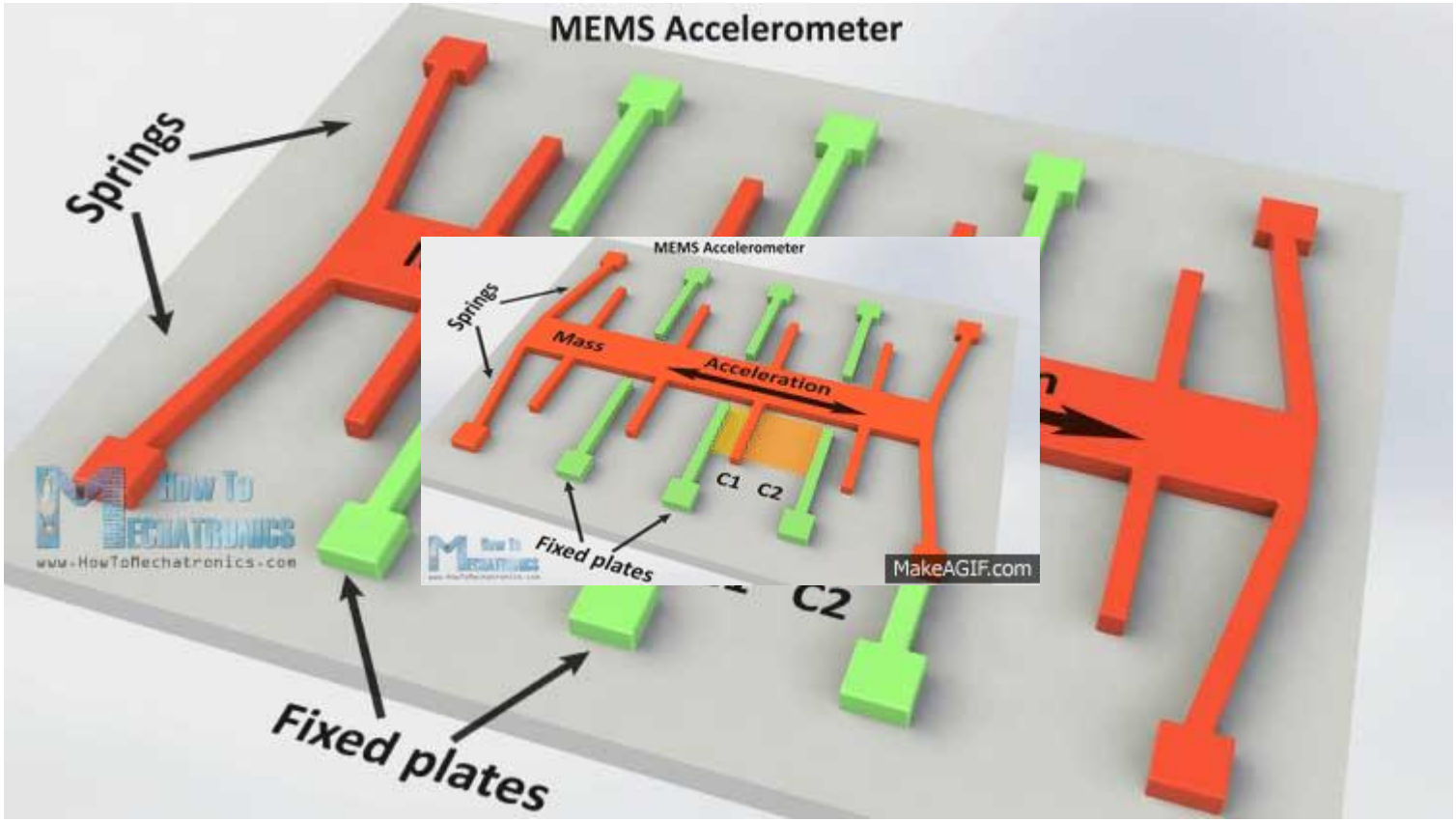

Mixed→Analog-Digital/Digital-Analog Converters

# Programmable Circuit Design

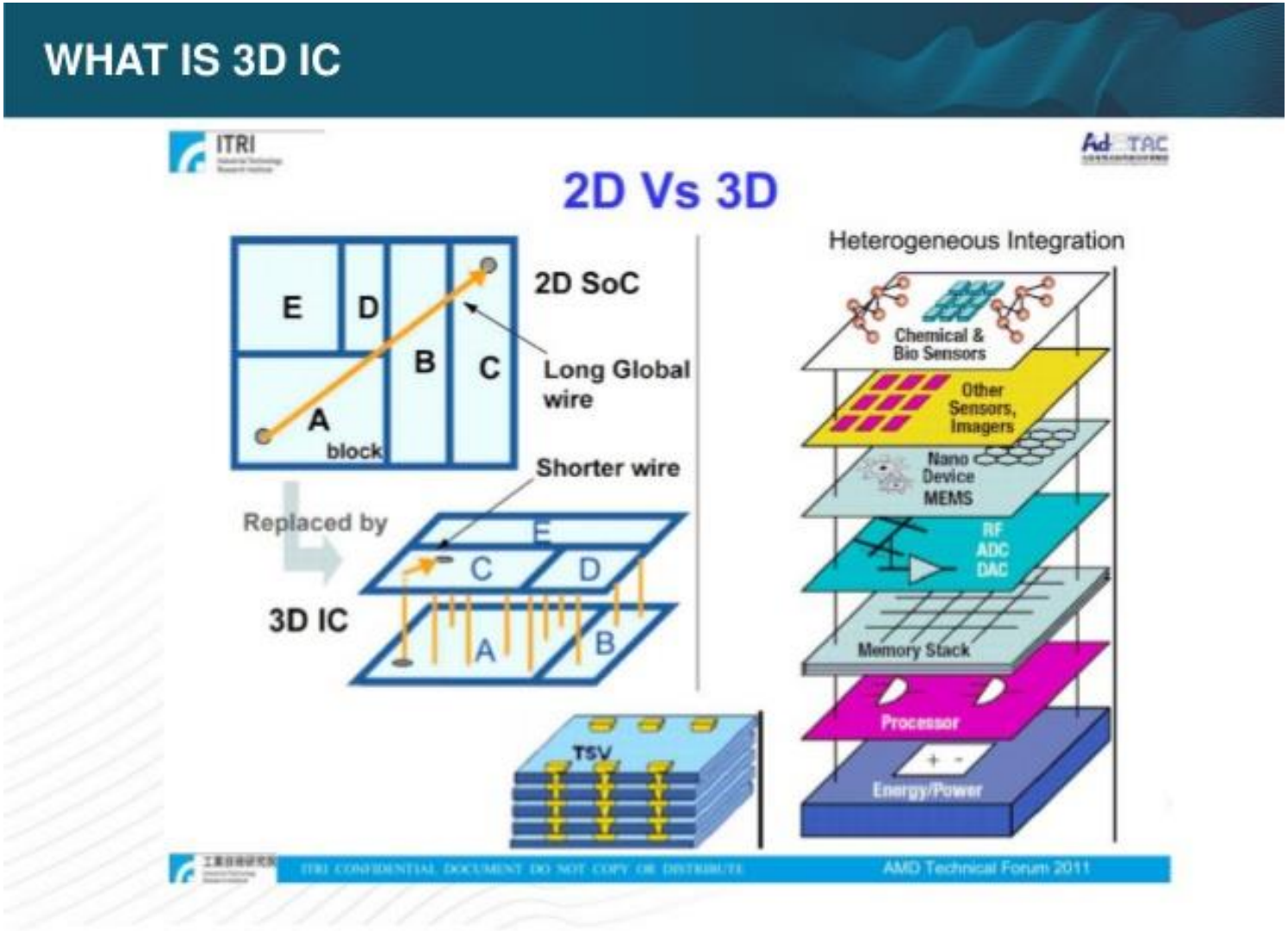
- PLA, CPLD, ve FPGA…



# MEMs- Microelectromechanical Systems



# 2D/3D Integration (MEMs+IC)

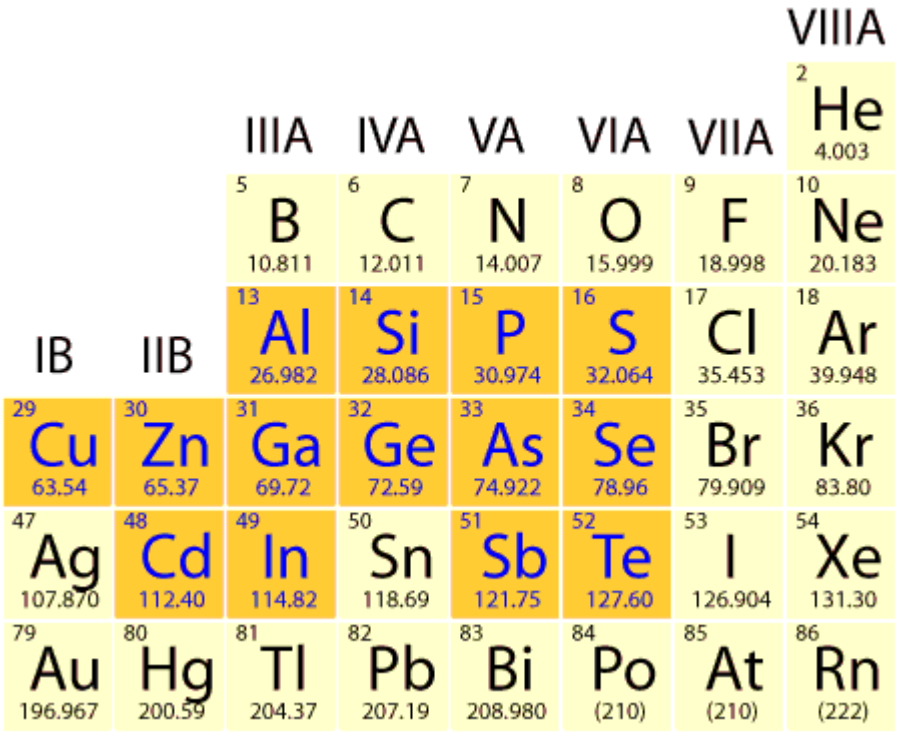


# Semiconductor Technology

- **Materials whose conductivity lie between conductors and insulators.**

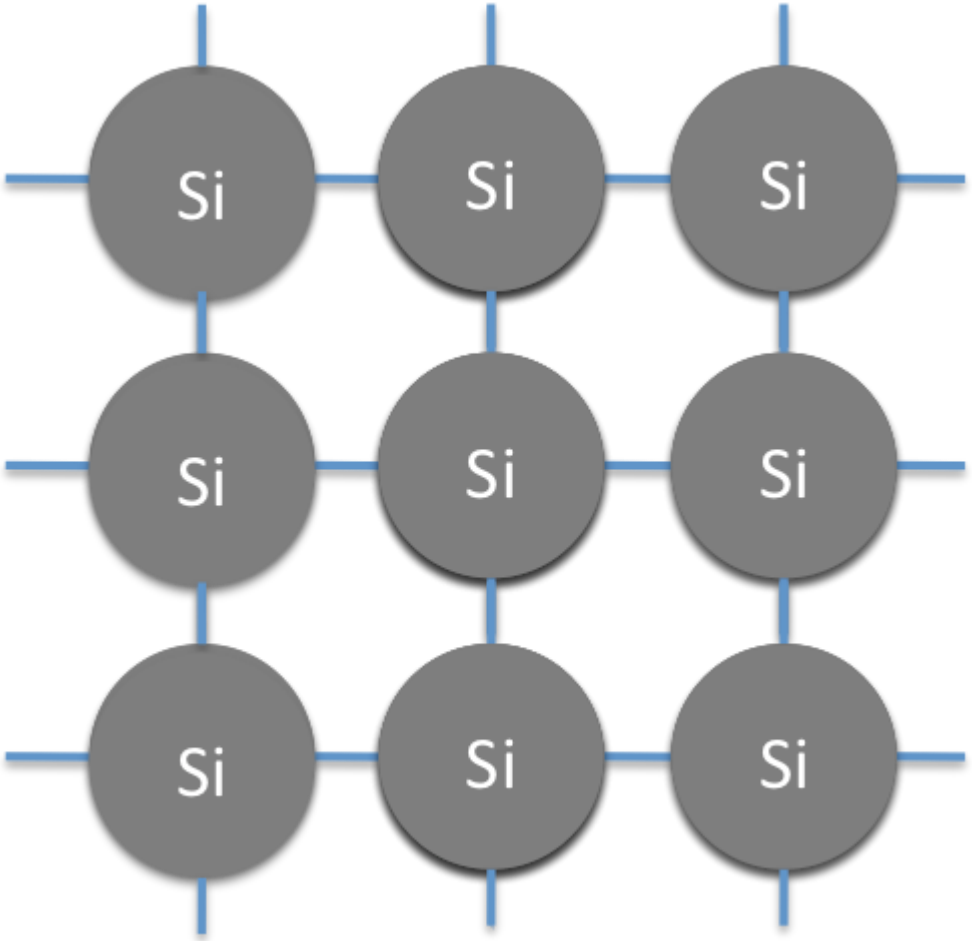
## Grup IV

- Si or Ge
- GaAs, InP, CdTe
- Si-Ge, Al-Ga-As

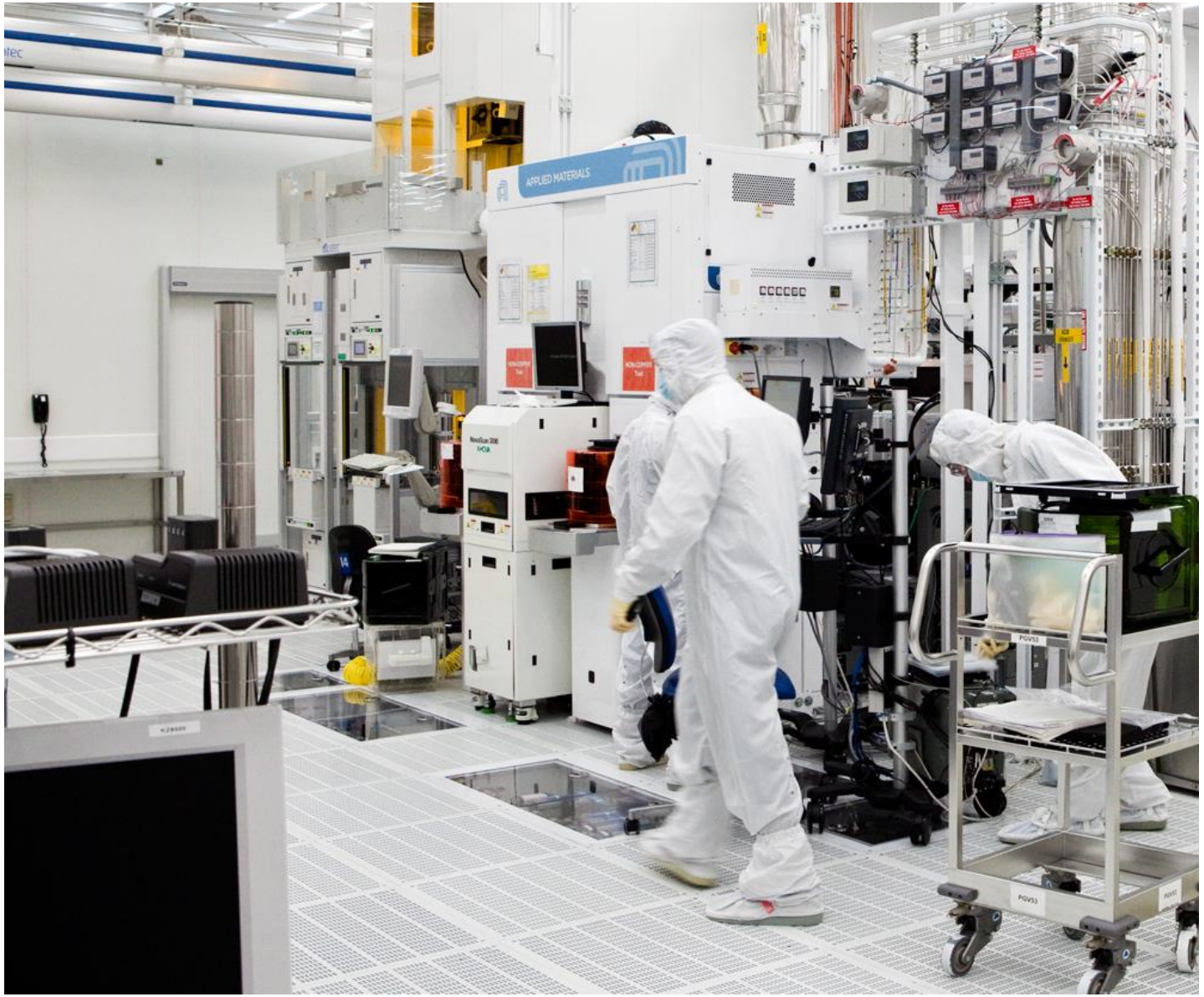


# Semiconductor: Silicon



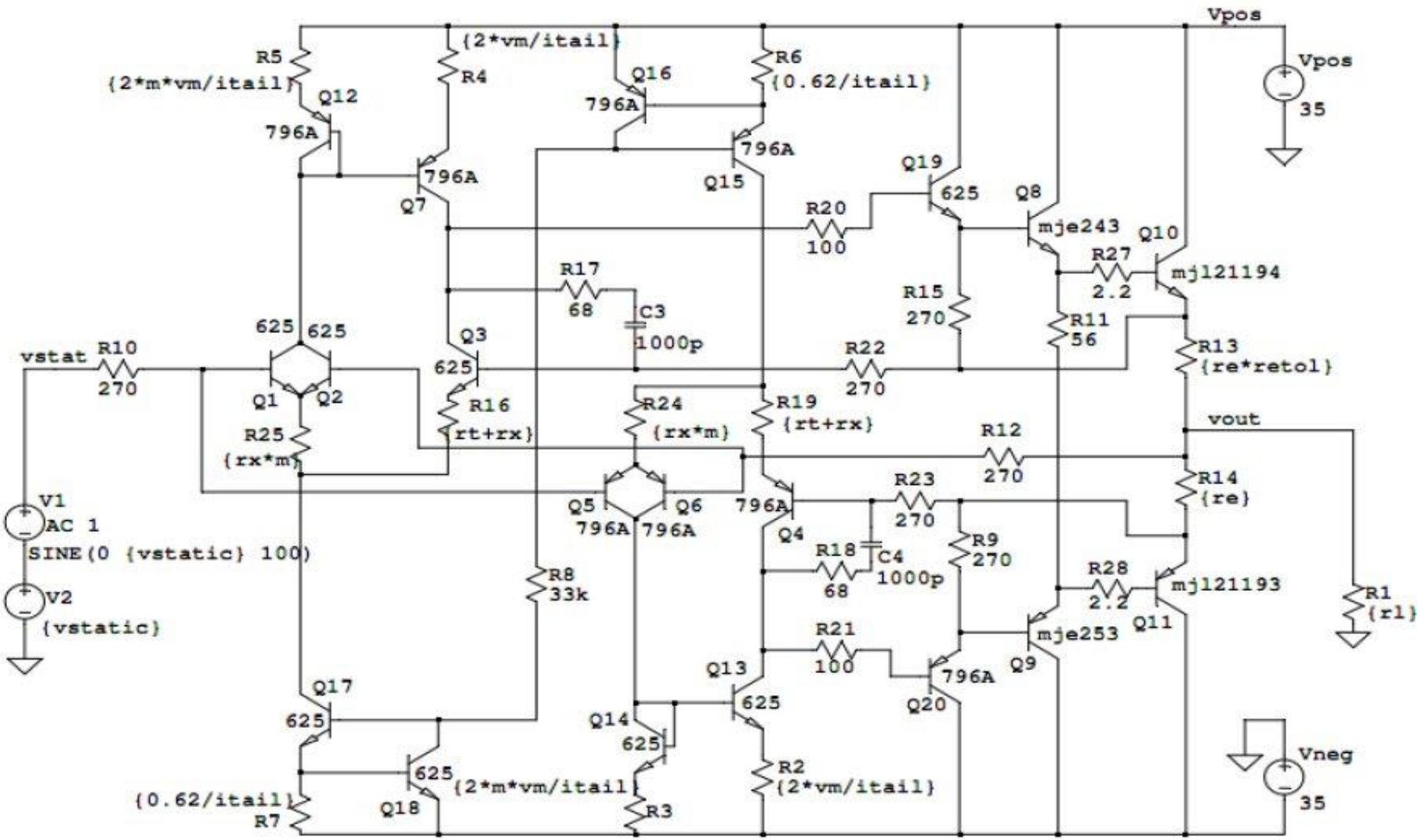
# Silicon Technology



# Silicon Circuit Technologies

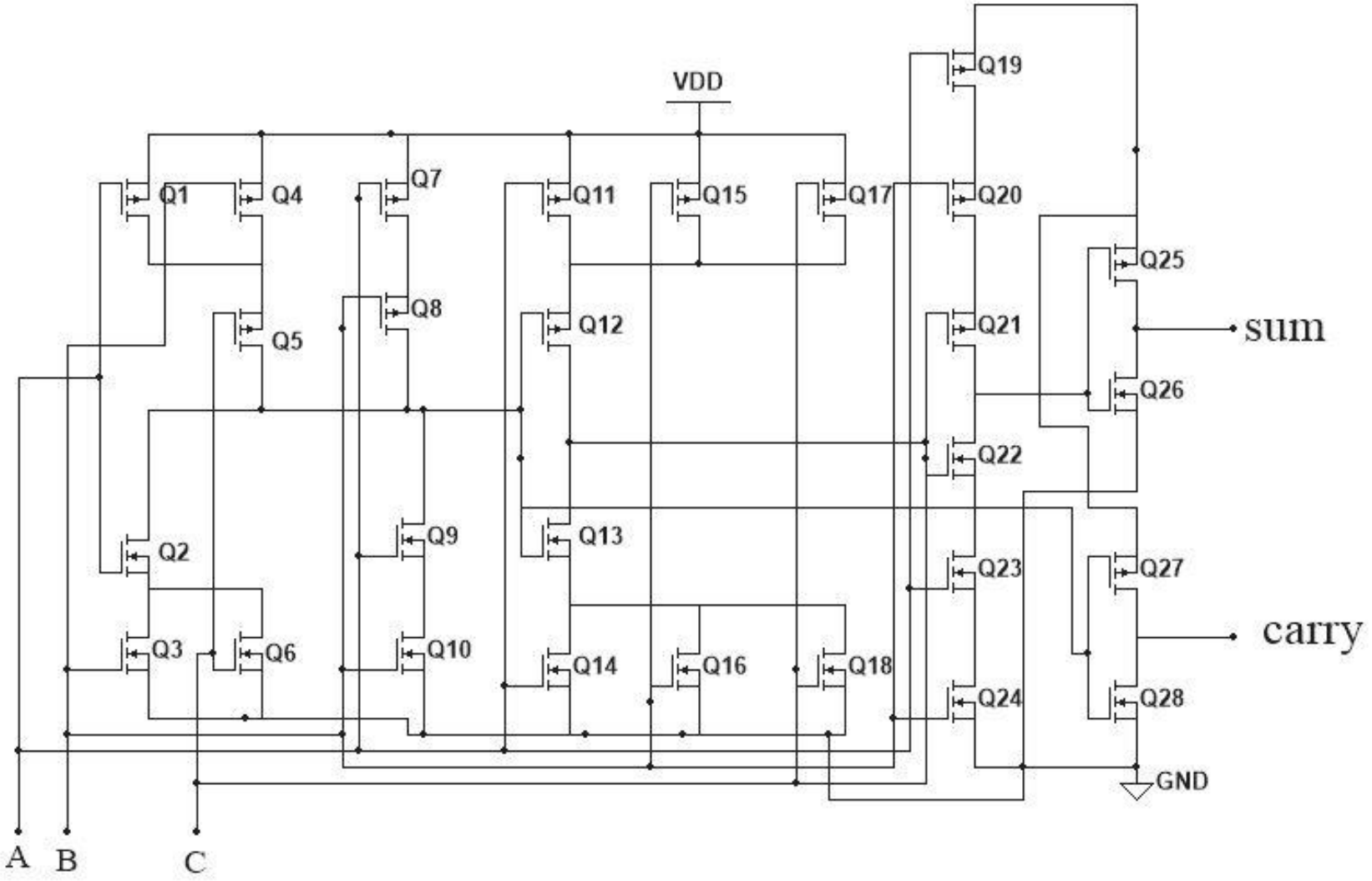
- **Bipolar Junction Transistors (BJTs)**
- **Comlementary Metal Oxide Semiconductor (CMOS)**
- **BiCMOS (BJT+CMOS)**

# Bipolar Junction Transistors (BJTs)




NPN devices marked 625 are FMMT625
PNP devices marked 796A are FZT796A
m, vm, itail, rx, rt, re and retol are adjustable parameters

# CMOS Technology



# BJT or CMOS ?

## Bipolar Technology

- High Switching Activity
- High current capacity and high gain
- Better noise performance with high frequency response

- High power consumption
- Low input impedance
- Low voltage swing

## CMOS Technology

- Low Power Consumption
- High density
- High Yield
- Low Cost
- Variable threshold voltage
- High input impedance

- Low gain
- Low current capacity
- Low switching

## BiCMOS Technology

**BiCMOS possesses superiorities of those.**

# BiCMOS Technology

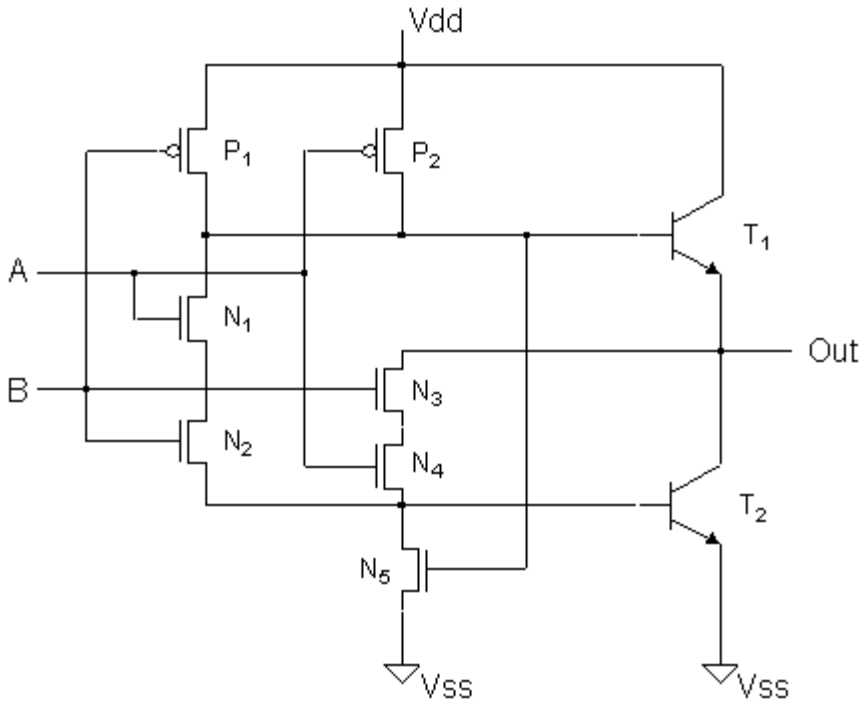



**Figure 8.1: The BICMOS nand gate circuit.**

# Integrated Circuit Design-How



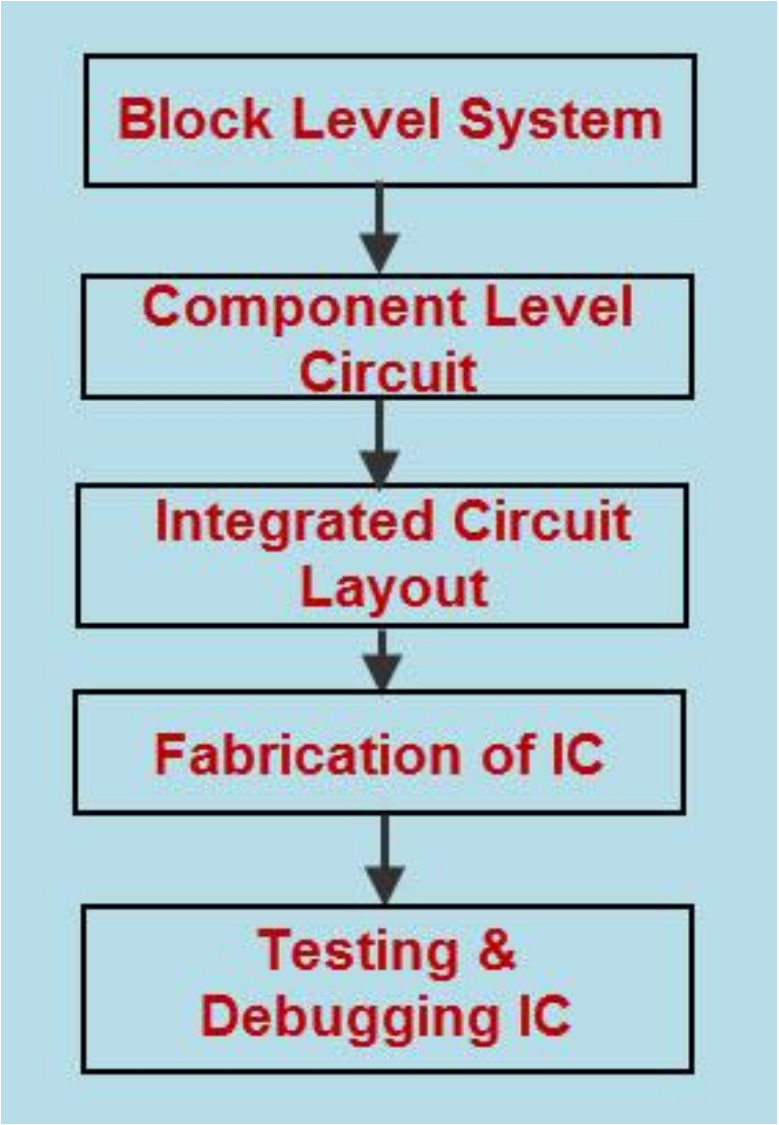









# Integrated Circuit Design-How




cadence™

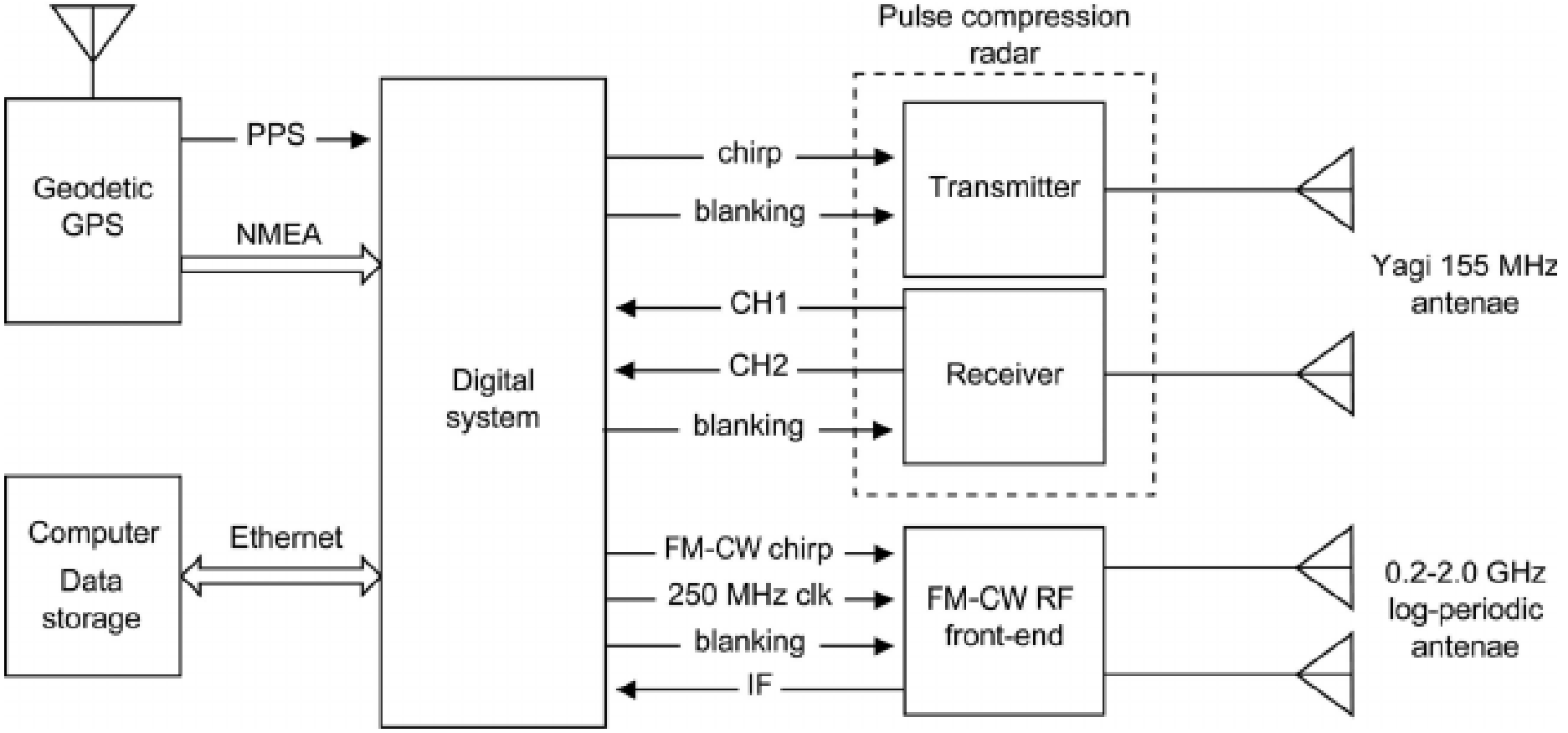
SYNOPSYS®

Silicon to Software™

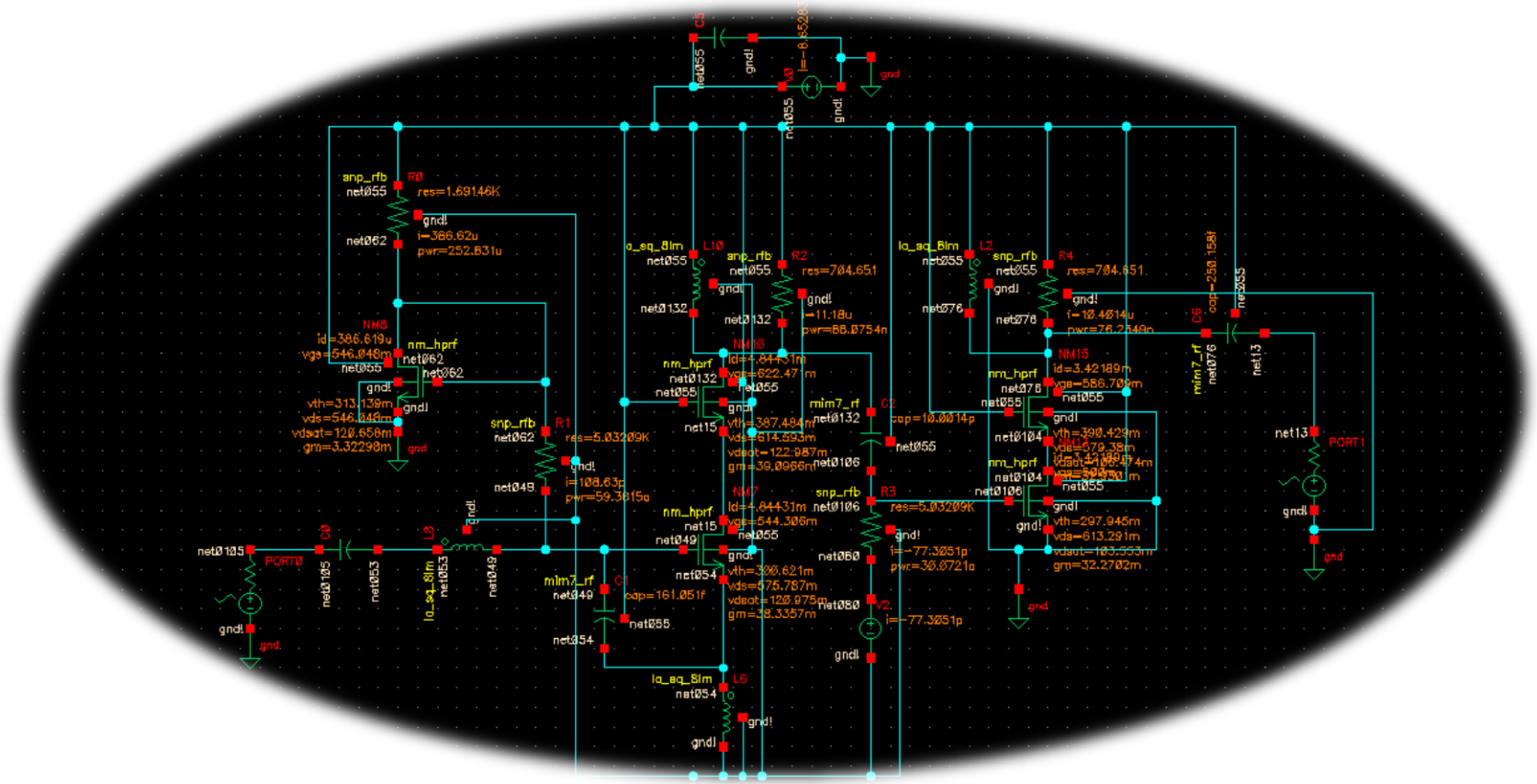
Mentor Graphics®

Tanner EDA

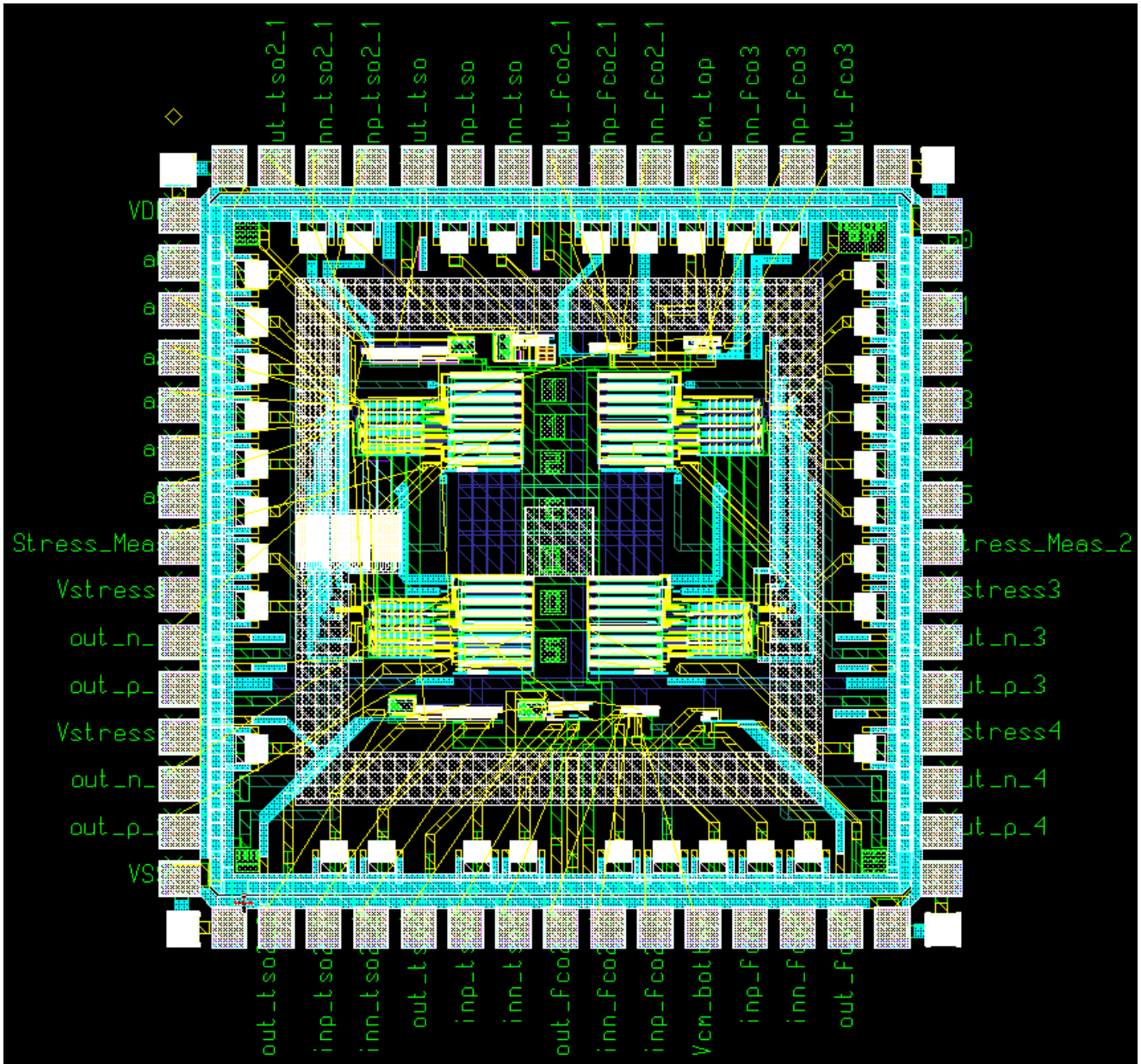
# System Level Design



# Circuit Level Design



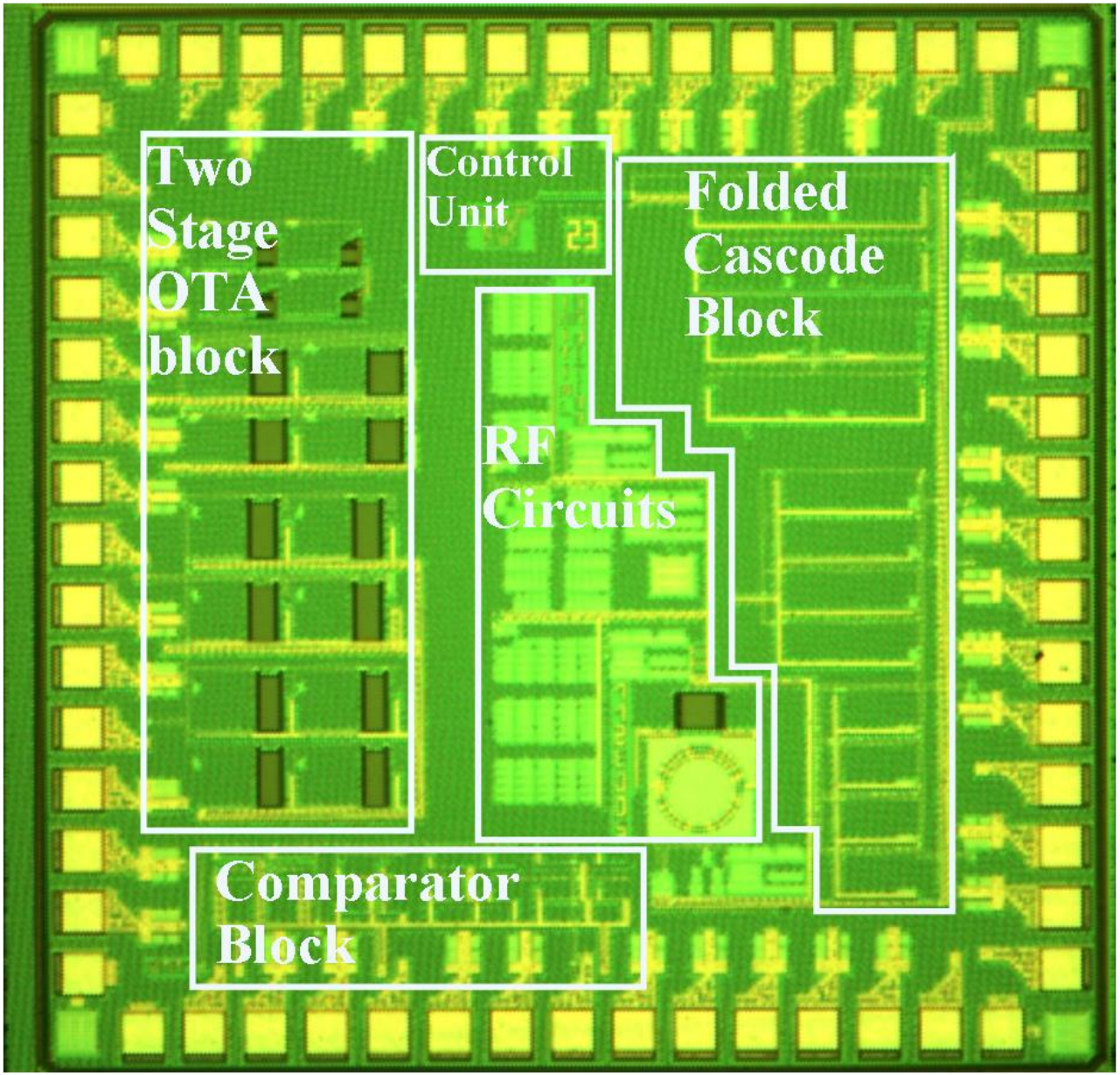
# Layout Level



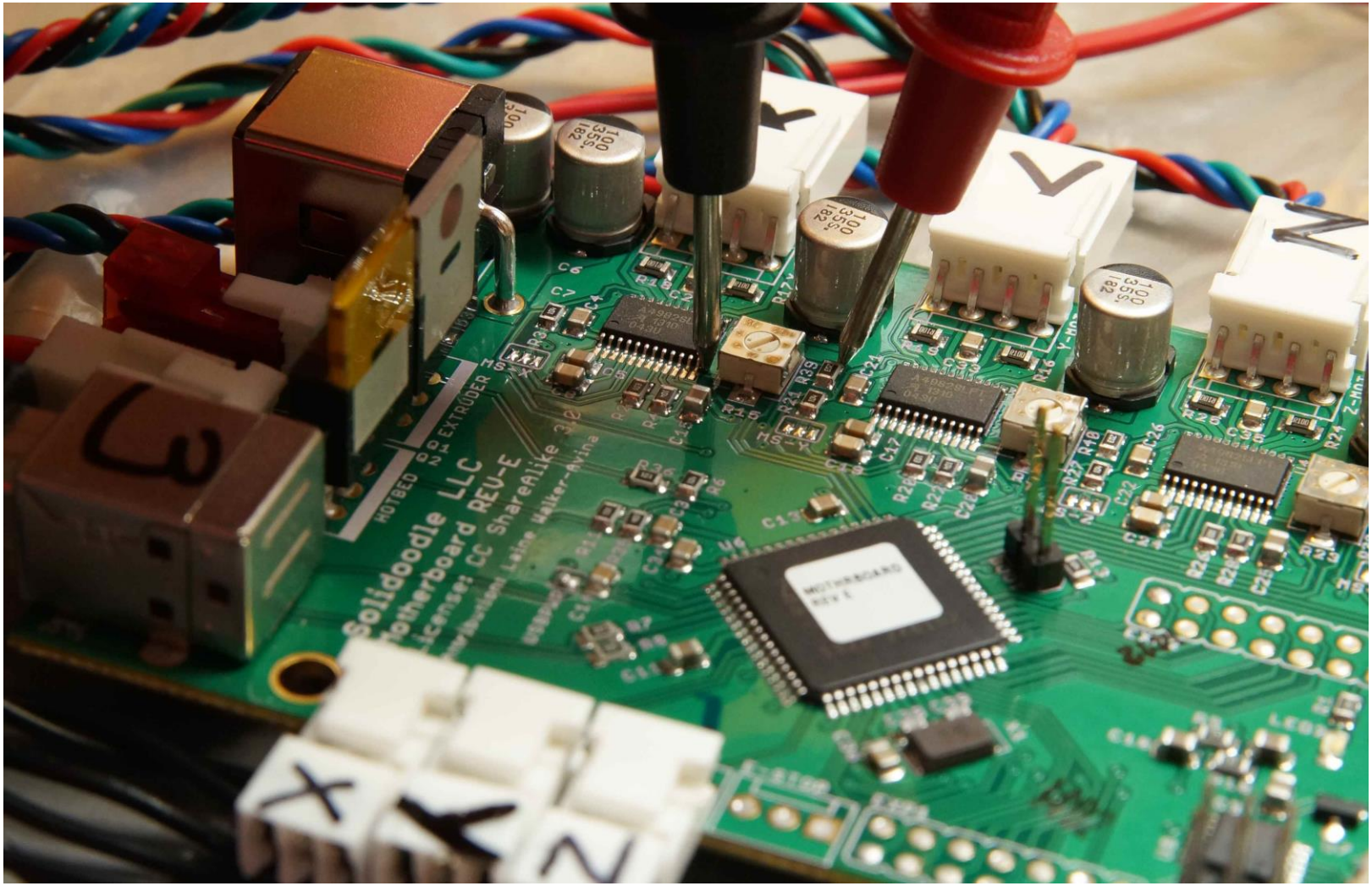
# Tape-Out (Production)



# Post-Production



# Test




TO BE
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